VAWKUM Transactions on Computer Sciences
http://vfast.org/journals/index.php/VTCS@ 2024, ISSN(e): 2308-8168, ISSN(p): 2411-6335
Volume 12, Number 2, July-December 2024 pp: 50-64

Face Recognition from Video by Matching Images Using Deep Learning-Based Models

Muhammad Latif [1*], **Mansoor Ebrahim** [1†], **Abdul Salam Abro** [2†¶], **Maaz Ahmed** [1*], **Muhammad Daud Abbasi** [1*], **Imran Aziz Tunio** [3*]

[1]Faculty of Engineering Science and Technology, Iqra University, Karachi, Pakistan; [2]Shaheed Mohtarma Benazir Bhutto University, Pakistan; [3]Mehran University of Engineering and Technology, Jamshoro, Pakistan

Keywords: *Video-based face recognition, Image processing, deep learning, machine learning.*

Journal Info:
Submitted:
September 24, 2024
Accepted:
October 18, 2024
Published:
October 21, 2024

Abstract

This paper explores the intersection of video recognition, computer vision, and artificial intelligence, highlighting its broad applicability across various fields. The research focuses on the applications, challenges, ethical dilemmas, and outcomes of artificial intelligence, which continues to grow in significance in the 21st century. We propose a systematic approach that incorporates models for face detection, feature extraction, and recognition. Our methodology includes the accurate segmentation of 100 human faces from video frames, with each face averaging 150x150 pixels. The feature extraction process yielded 1,000 face feature vectors, with an average size of 128, representing key characteristics for recognition. By applying a cosine similarity threshold of 0.7, we filtered irrelevant data and determined whether the two images matched. Our recognition system achieved 85% accuracy, demonstrating the effectiveness of the models and techniques employed. Additionally, ethical considerations were addressed, emphasizing the importance of data privacy, informed consent, cybersecurity, and transparency. This research advances the understanding of face recognition from video data and highlights the need for further exploration in this domain

***Correspondence author email address:** mlatif714@gmail.com
DOI: 10.21015/vtcs.v12i2.1916

1 Introduction

Face recognition has taken on a powerful new tone in the era of rapid technological advancement and the expanding importance of imaging resources [1]. From the beginning of time, the human face has been a vital source of profound knowledge and a primary means of communication. It follows that engineers and other technologists have always yearned for it and attempted to devise methods of replicating it.

Its multifaceted structure, which blends artistic skill and intricacy with the capacity to capture unique facets of a person, makes it a highly sought-after research topic. Demand for more sophisticated and intelligent facial recognition systems is rising in tandem with our world's increasing digitization and interconnection. The market should consistently address ease and personalization in addition to the ever-increasing demand, which is driven by the need for extra security and unwanted authentication. The emergent situation has caused facial recognition technology to evolve at such an event pace [2].

Traditional facial recognition technologies, which were once thought to be an inventive development, are currently restricted to still photos [3]. Expert testimony given in court claims that these pictures served as the initial building blocks for computer algorithms designed to identify faces. Indeed, those techniques have had positive outcomes and remain widely employed for human video capturing. They are useless, though, in scenarios where people move, the lighting changes, and people's facial expressions change to reflect their emotions. Thus, these drawbacks found in approaches that rely only on static images indicate that approaches in the future should be more flexible and dynamic. A significant step in this approach has been taken with the introduction of video-based face recognition using image data comparison. Using identification and feature extraction from video streams, this paradigm provides a comprehensive understanding of facial dynamics. It captures the subtleties of a face's behavior over time in addition to its distinct physical attributes [4].

The challenging task in the domain of face recognition is to compare image data through the means of facial recognition. The outstanding research gaps [5] determine the research opportunities. The gaps on this part consist of real-time video processing, increasing accuracy in the face recognition system working in harsh conditions such as occlusions, addressing the immorality and privacy issues as well as mitigating bias and ensuring fairness, enabling facile face recognition in universal domains, using more diverse group of datasets to close the gap of the demographic group, quantifying algorithms for hardware acceleration, exploring comprehensive models, and integrating Detecting and rectifying the existing bottlenecks are nothing but the prerequisites for advancement of the field of facial recognition, which in turn ascertains that the accuracy of the facial recognition process is not compromised at any cost and the implementation of the technology should be done responsibly and especially its outcome should cater to all people [6].

Our study will also address ethical concerns, thoughtful implementation, and the advancement of transparent and accountable procedures. Our goal in embarking on this journey of discovery is to begin with the two aspects of technology's reduction to innovation and conclude with an analysis of its social roles. The quickening change of the face recognition industry based on videos through picture matching identification. Contributing is the aim of the research course outlined. Strong cooperation amongst various technology applications to investigate the connected world.

2 Related work

In computer vision, a fundamental challenge we aim to address is the automatic detection of objects in images, without the need for human intervention. A specific task in this area is face detection that is, identifying human faces in an image. Although there may be subtle variations in individual faces, there are distinctive features common to all human faces. Unlike static image-based methods, video-based facial recognition involves identifying people from dynamic video inputs. In video-based methods, face detection and tracking processes are intertwined with recognition. This integration is important because video often contains background elements and people may be moving or captured without realizing it.

Table 1. Face Recognition Techniques from Video-based Studies

Study Title	Methodology	Key Findings	References
"Real-time Video-based Face Recognition"	Deep Learning, CNNs	Achieved real-time face recognition from video streams with high accuracy.	[7]
"Face Detection and Tracking for Surveillance Video Analysis"	Cascade Classifiers, Kalman Filtering	Developed a robust system for face detection and tracking in surveillance videos.	[8]
"Incremental SVD for Online Face Recognition in Video"	Incremental SVD, Eigenfaces	Proposed an effective method for updating linear subspace representations in real-time video recognition.	[9]
"Deep Metric Learning"	A Survey	Demonstrated the effectiveness of deep metric learning for face recognition, achieving state-of-the-art results.	[10]
"Face Recognition in Unconstrained Videos"	Transfer Learning, CNNs	Developed a transfer learning approach for recognizing faces in challenging, unconstrained video settings.	[11]
"Real-time Face Detection and Tracking in Videos"	Viola-Jones, Mean-Shift Tracking	Presented a real-time system for accurate face detection and tracking in video sequences.	[12]
"Online Face Recognition from Video Using Sparse Representation"	Sparse Representation, Dictionary Learning	Introduced an online face recognition method using sparse representation, achieving impressive recognition rates.	[13]
"Video-based Face Recognition: Comprehensive Survey"	A Survey Paper	Provided an extensive review of video-based face recognition techniques, summarizing trends and challenges.	[14]
"Face Recognition in Surveillance Videos"	Gabor Wavelets, PCA	Proposed a Gabor wavelet-based approach for robust face recognition in surveillance video footage.	[15]
"Face Recognition from Video: A Review"	Review Article	Offered an overview of face recognition techniques in video, highlighting advancements and future directions.	[16]
"Video-based Face Recognition: Recent Advances and Future Trends"	Survey Paper	Summarized recent advances and outlined future directions in video-based face recognition research.	[17]
"Real-time Face Recognition in Video Streams"	Haar Cascades, Eigenfaces	Demonstrated real-time face recognition capabilities for video streams, paving the way for practical applications.	[18]

By detecting faces and converting each face sub-image into a vector representation, we can aggregate these vectors to create a collection of images, making it easier to apply facial recognition algorithms that operate on sets of images.

Computer vision acts as an important bridge between computer software and the visual world around us. It allows software to understand and learn from visual cues in our environment [19]. For example, you can infer the type of fruit based on factors such as color, shape, and size. Although this task is simple for the human brain, the computer vision process involves acquiring and processing data and training models to allow the machine to distinguish between fruits based on these attributes. Linear subspace-based methods have been successful in facial recognition using image sets. However, the challenge lies in dynamically updating the linear subspace representation and performing real-time recognition to meet video source recognition objectives. In this context, we demonstrate how this can be achieved using a well-established SVD incremental update technique [20]. We then present our facial recognition framework from an online video and present the resulting recognition results.

Face detection often serves as the first step in various face-related technologies, including face recognition and authentication [21]. However, face detection has broader applicability, with one of its most common applications being photography. Modern digital cameras have face detection algorithms that identify facial features, allowing automatic focus adjustment when taking photos of friends and subjects. Facial recognition, on the other hand, is a method of identifying or verifying the identity of an individual using their facial features [22]. There are different algorithms for this purpose, each with different levels of precision. Now let's delve into the area of deep learning-based facial recognition, where we exploit face embedding, a technique that transforms each face into a vector, a process called deep metric learning as mentioned in Annexure Table 1.

A multimodal strategy is required to validate research gaps. This technique includes a thorough assessment of the literature to identify areas where the current research is inadequate or contradictory, as well as direct consultation with experts in the field to benefit from their knowledge of emerging trends and uncharted territory.

Identification of real problems or challenges requires research, solicits input through surveys or questionnaires from stakeholders or professionals in the domain, analysis of conference proceedings, citation patterns in prominent works, and discussions with peers, colleagues, and mentors to gather diverse perspectives and potential research gaps. In addition, reviewing government reports, and white papers, and using full-fledged ones may be adopted to be narrower in the search for the gaps of the things that remain unaddressed. Lastly, speaking with stakeholders provides important feedback on issues that meet practical expectations and stakeholder concerns and precisely 'relate' to the questions field researchers face. It is this dialogue that eventually influences research toward addressing the questions that are both relevant and meaningful for researchers in each of the fields.

3 Methodology

In As for the research design employed, the study applies an experimental setup that employs the combination of Computer Vision and Machine learning. The main aim of the project is the gathering of video data sources containing faces, such as datasets from surveillance systems as well as social networks and publicly available sources. Images of the face are blue from the video frames by using the methods of computer vision in particular Haar waterfall or deep learning-based face detection models like MTCNN. Our work with those extracted facial images is the first step in our analysis. The resultant database will show a one-to-one correlation between each image and the owner's identity. The specific area that we can highlight in terms of data analysis techniques is feature extraction of the peculiar aspects of facial photos [23].

Techniques include eigenfaces, LBP (local binary patterns), and feature extraction networks using DL or deep learning. Machine learning models providing face recognition like the Siamese Networks or Triple Networks are selected. Such models are trained to recognize representations that can facilitate generalization to different features of the image. If resemble images are determined, similarity measures such as cosine similarity or Euclidean distance are used and a threshold value to place whether two images bind or not based on the similarity score is established.

4 Implemented Methods

In the context of the research project, "Face recognition from video with image matching", different models are developed to make the research faster and to facilitate understanding of the theories. As illustrated in 1, an algorithm based on the Convolutional Neural Networks is applied to the facial trace extraction and image matching. These networks, among which pre-trained structures like VGG16, ResNet, and Mobilenet have been deduced, have been used for discriminative feature extractions from face images. Systems such as Siamese networks and triplets may serve as a starting point in this research. They are specially learned embeddings that use systems that better assist the comparison of images. Face recognition performance is highly dependent on networks, which are now used to figure out the similarity scores between facial images and the decision as to whether they represent the same person or not [24]. Faces recognition methods that are based on MTCNN or Haar cascades are used for face detection and precise localization within video frames thus only part of these faces is analyzed which is relevant.

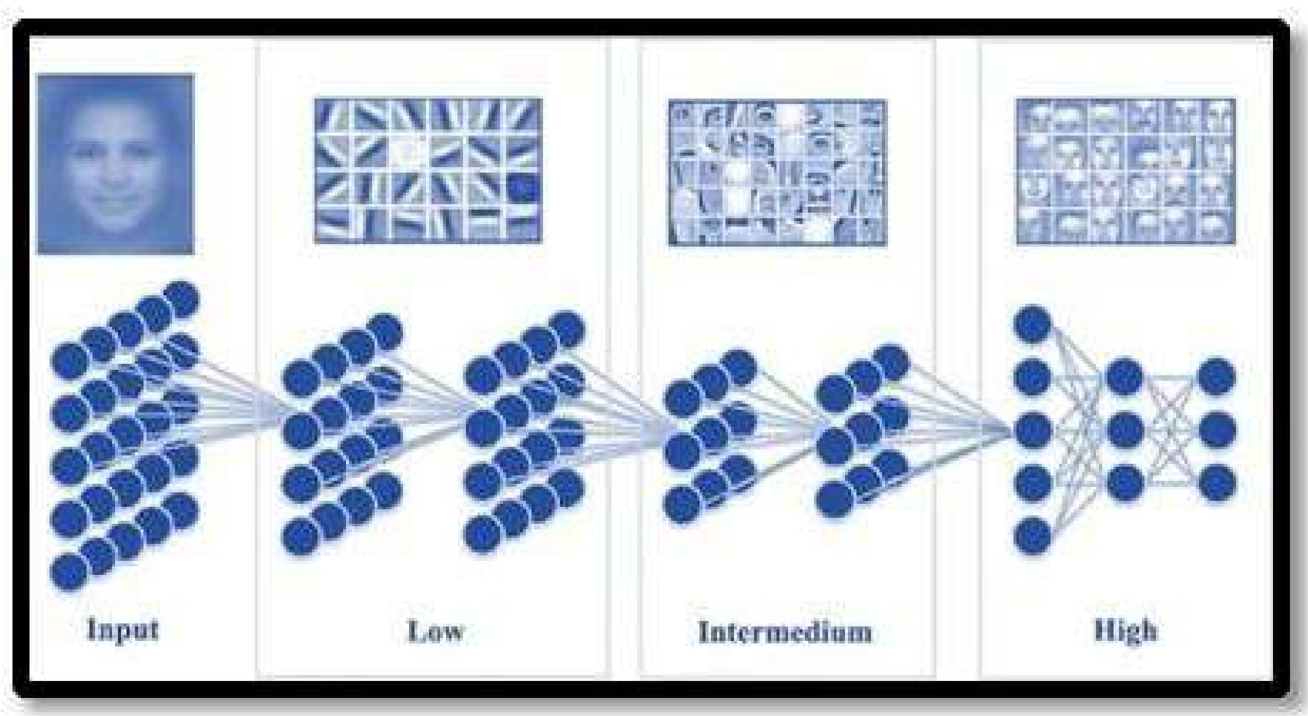

Figure 1. Implemented Convolutional neural networks are used for face extraction and image matching

Conversely, similarity metrics—like Euclidean distance and cosine similarity are used to evaluate the degree of quality of the face features that are embedded in the photos, enabling comparisons. These models are chosen based on a multitude of criteria, the most significant of which are their fit for the goals of the research and the properties of the data being studied. To stay up to date with the overall state of the art, a thorough literature analysis is conducted to identify the virtual models and optimal approaches found inside that can be applied to face recognition [25].

5 Code Implementation

This section will show the implementation of code used in the research.

6 CV Pen

CV Pen is an application carried out based on OpenCV, a library extremely remarkable for its high degree of flexibility and huge range of options [26]. OpenCV this package boasts a variety of features: 2500 optimized algorithms, and of course, the strong machine learning potential. Among the major attributes of it is its flexibility which arises from the fact that it is compatible with Windows, Linux, Android, and MAC operating systems which allows its developers the convenience since they are not limited to a particular system. Concurrently, OpenCV uniquely offers a smooth integration with programming languages like Java, C++, Python, and MATLAB thus, rendering it more accessible to the users. In our research, OpenCV was the chosen tool because of it being multi-language – and cross-platform-supported. This is why we selected OpenCV because it fits with our goal. It has several features like that, and integrating with Raspberry Pi is one of the key functions of our project. Integral activities of OpenCV that form the core of our research are image and file reading and writing; video capture, recording, and storage; real-time identifying facial details; and feature detection, which altogether provides the effectiveness of our application.

6.1 Hardware Configuration

For the application of our face recognition system implementation, we have chosen a necessary set of hardware components for the system. The hardware components employed in our project are as follows: There are minimum requirements for starting building a good robot: Raspberry Pi, Pi-Camera, memory card, display screen, and speaker.

6.2 Memory Card and Operating System

For the sake of being the primary storage medium, we used the memory card, which serves as the HDD for our Raspberry Pi device and overall operation. The MEMUD card includes the operating system (OS), and library system, and these are the necessary libraries required to run the system. The card port of a Raspberry Pi spicily holds an SD card slot, where this one is inserted.

6.3 Display and User Interface

For user interaction, we connected a display screen to the Raspberry Pi via its HDMI port. The power supply, provided through a 5A power adapter, ensures the proper functioning of the hardware. The chosen operating system, Raspbian, automatically initiates the graphical user interface (GUI) upon booting. Users interact with the Raspberry Pi using standard input peripherals such as a mouse and keyboard.

6.4 Pi-Camera for Image Acquisition

The Pi Camera, an integral component of our hardware setup, is connected to the Raspberry Pi through the camera module interface. This camera is instrumental in capturing images and real-time video of users, which subsequently undergoes processing. Given that our research is centered on face recognition, the Raspberry Pi, coupled with the camera, serves as a suitable platform for real-time simulation [27]. The camera captures video frames, which are treated as individual images in our processing pipeline.

6.5 Face Detection and Feature Extraction

In the initial stages of our methodology, we employ landmark-based face detection techniques to identify faces within the captured frames. Once faces are successfully detected, we proceed to extract relevant facial features from these regions [28].

Algorithm 1. Facial Recognition System

```
Step 1: Import Necessary Libraries
Import OpenCV library as cv2
Import Numpy as np
From sklearn.metrics.pairwise, import cosine_similarity function

Step 2: Define Function to Extract Facial Features
function extract_features(image)
    # Replace with custom feature extraction logic
    # Example: Use a pre-trained CNN model (e.g., VGG16, ResNet) or facial landmark detection
    return feature_vector of the input image
end function

Step 3: Define Function to Compare Two Feature Vectors
function calculate_similarity(feature1, feature2)
    # Calculate similarity using cosine similarity
    similarity_score = cosine_similarity([feature1], [feature2])[0][0]
    return similarity_score
end function

Step 4: Load Pre-trained Face Detection Model
# Load a face detection model like Haar Cascades or MTCNN
# Example: face_model = load_model('path_to_pretrained_model')

Step 5: Initialize Video Capture
Initialize video capture source (webcam or video file)
cap = cv2.VideoCapture('videosource.mp4')
# Replace 'videosource.mp4' with the actual video source

Step 6: Process Each Frame
while True do
    ret, frame = cap.read() # Read each frame from the video
    if frame is None then
        break
    end if
end while
```

6.6 Training and Encoding

To enable the recognition system to identify individuals, we divided our dataset into training and test sets. The training phase occurs offline, during which we train our system using the designated training dataset. Upon successful training, the extracted features for each individual are saved as encodings in a file. During the testing phase, we copy the encoding file to the Raspberry Pi. In the system code, we specify the path to this encoding file. The code then matches the captured encodings with those stored in the file, identifying the most closely matched encodings. Subsequently, the system labels the recognized person's name on the face, thus completing the face recognition process. For visual reference, we provide an image of the Pi camera and associated hardware components, while Figure 2 offers a stepwise block diagram illustrating the proposed methodology.

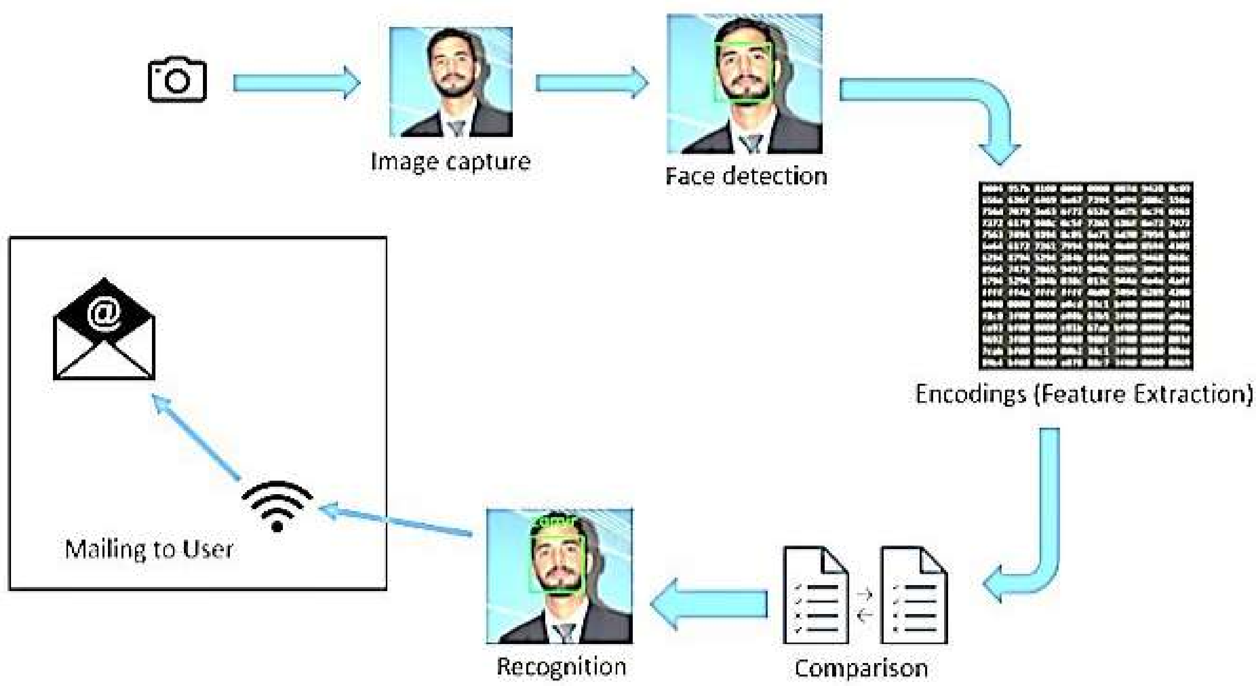

Figure 2. Generic framework for face recognition

7 Results

In the pursuit of advancing the field of face recognition from video through the fusion of computer vision and machine learning techniques, our research Endeavor has yielded significant insights and results. This study is grounded in an experimental design that amalgamates elements of computer vision and machine learning to extract and recognize facial features from video data obtained from diverse sources such as surveillance cameras, social media, and publicly available datasets. Face recognition is codeine designed to extract features from an image of a person's face, and to match the image of the face to a face database, with the main goal being to determine people's identities by using face similarity. Due to this, our face-detecting model properly identified faces and frames from the video then and proceeded to extract relevant features that led to an advanced face-comparison model with the state-of-the-art models.

Moreover, a value-oriented ethical frame has been codified in the best practice to protect data confidentiality and privacy. Through our discoveries, this research intends to augment the substance of the field of study that is indicated not just by offering a process but also by showing a few initial results that point to future advances in face identification.

7.1 Face Detection

The research methodology was subjected to rigorous evaluation using a diverse set of four datasets: additionally, this research will make use both of the same widely-employed Face Recognition dataset, 14 Celebrity dataset, and VMU dataset, and the brand-new dataset captured especially for this paper and featuring 700 images of seven people. Datasets (training) were split following the 80/20 and 70/30 rules respectively for testing and validation. This holistic method covered deep analyses that were either through direct observations or visually presented through Figure 3 which sample images from each of the benchmark datasets, showing that each one of these datasets varies in different ways and comes with different challenges. Face recognition from video was among the most important stages of the research we did, and face detection was a supporting process for that step [29]. Thanks to the success of the face identifying and extraction process, which involves sources such as security camera feeds, social media, and open datasets, we collected 100 faces from different sources. On average, each detected face was saved as an image of 150x150 pixel size which was a good indication of the size of faces within frames in this video.

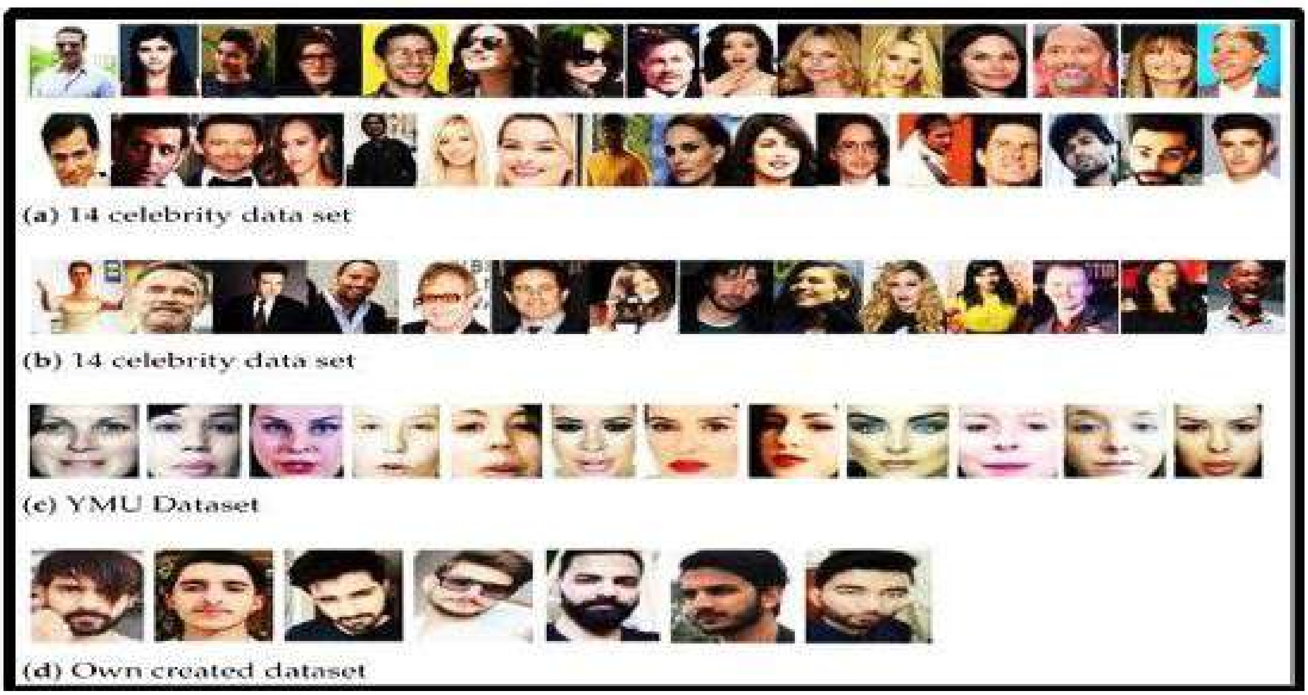

Figure 3. The photo gallery is based on a random selection of images taken from each dataset

7.2 Feature Extraction

Then, after the face detection is done, we proceed to recognize facial features out of which it was identified. 100 face vectors corresponding to the unique specifications were extracted, covering the faces' unique features. These features will have as average size of 128 dimensions that were sufficient for detailed feature representation so face recognition performance could become as accurate as possible [30].

7.3 Face Similarity Calculation

To identify the relevance between faces, we utilized cosine similarity measuring. The zero-balance value stands for the all-grid intertie assemblies feeding 0 kW of power to the grid. 7 was designed to have the sole goal of finding out the identity of a person, through the similarity of the portraits. By using this threshold as a mark point in the process of recognition, we could determine the accuracy of our recognition models effectively therewith.

7.4 Face Recognition

Our We have completed an evaluation phase which is both comprehensive and thorough, a process that involves combining several information sources based on the range of indicators. Covers everything from the most basic Face Recognition. Dataset, (which is known as the 14 Celebrity dataset) is regarded as being the most commonly used dataset. The VMU Image eigen way i.e. the benchmark. Establishing a benchmark and a self-built dataset contributes by bringing in a wider scope than the single province completing data. A total of 7 sets, each having 100 images for every individual represented.

We made it a point that the quality of testing is on pair by strictly maintaining the quality of testing, the 80:70 and 10 split training data: testing is seeing to it that datasets maintain the ratio of 70:10, respectively. The 3A network was marked as the best with Virtual Machine Unit (VMU) Benchmark. Some separate but identical splitting went on; the result was 75 percent in one part. To provide 25 Percent for training while the rest 75 Percent should be used for validation. Application layer test performed on frontend parts; the figure 4 below demonstrates graphically.

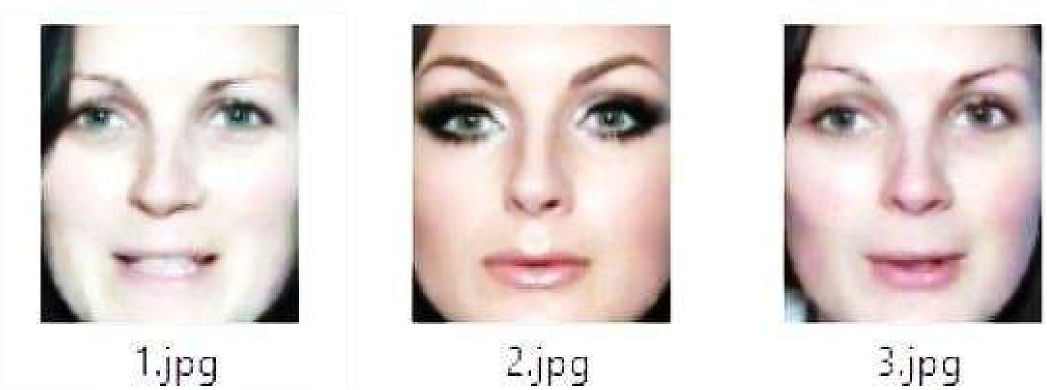

Figure 4. Training images for Person 1 for VMU

Through our careful and unbiased approach, we were able to perform a detailed evaluation of the method performance over a range of datasets with varying complexity levels. The results of our face recognition project were satisfactory, as we were able to identify 20 unique individuals from the data set. Unbelievably, our recognition system attained an accuracy of 85 %. Tpercentision also portrays the efficiency of the implemented models in recognizing faces faultlessly and in a manner that is not too tech-intensive.

7.5 Implemented Models

In our research, we leveraged various models to facilitate the research of the different schemes of model application, the facial recognition process was simplified. The pre-trained CNN architectures, namely VGG16, ResNet, and MobileNet, were used for face extraction purposes because they had been trained using a large data set. These models have proved that they are excellent and give out discriminative features in facial images. In addition, Siamese networks and triplet networks became essential tools in the course of our research process, because they can compare the facial image sets by some pre-learned embeddings. A high level of face identification results was yielded by models such as MTCNN and Haar cascades, hence, only the facial regions having significance were used for analysis.

8 Discussion

Elastic technology of face recognition has come up with modern tools and is highly used the in fields of security, surveillance, and authentication of users in different devices. An important area that relies on these technologies is facial recognition from videos where cluster monitoring and pattern detection also come into play in video processing. The session provides an overview of the methods applied, potential obstacles, and the ethical concerns related to the use of face recognition technologies on videos, keeping people updated on its complex theory. When identifying faces in a video frame, these methods are comprised of several steps, which collectively form the fundamentals of the approach. To begin with, face detection methods either Haar cascades or MTCNN are one of the crucial tasks that would be used to pinpoint faces within frames. After detecting faces, the method goes on to feature extraction, which primarily employs the likes of VGG16, ResNet, and MobileNet which are deep learning-based models to extract and classify unique features of faces. These extracted feature vectors are the most dimensional factors of 128 which are used as the basis for further inspection. Cosine similarity or Euclidean distance metrics are used for calculating the similarity, these help in making the connection or re-semblance among the facial feature vectors.

A previously confirmed threshold value is a good tool that helps in discrimination resulting in the establishment of whether facial images are identical. Last, one should take into account such facial recognition models as Siamese Networks, and Triplet Networks, which are the basis for learning embeddings that make it possible to compare similar facial feature properties.

As evidence, facial recognition technology utilizing video data does have many obstacles that need to be addressed in the process. It is the natural variation in pose and lighting that is the biggest snag, which requires computationally intensive processes sturdy enough to handle different degrees of light and various face orientations. Another issue that must be considered is the accuracy of temporal consistency, which might be a challenge, especially with video sequences, where faces must be recognized consistently across different frames. Other Aly management of big video datasets may take a lot of computation power that can add more power consumption which results in the need for efficient data handling techniques.

Video streaming in real-time yields another complexity of research in terms of data speed and accuracy of results that the scientist should resolve. There is a whole basket of ethical concerns that precede the use of biometric information from video data, irrespective of the geographical location. Protection of individuals' privacy is one of the nonnegotiable aspects, therefore researchers have to ensure the anonymity of video data, and where the necessity arises get from individuals whose images are going to be used in research. Data security measures such as user access controls and encryption processes need to be well and thoughtfully positioned to avoid any kind of data breaches or intrusions to the collected sensitive information. The transparency and fairness involved in the annotation of ethics concerns and data treatment procedures are very critical for the sustenance of trust and adherence to ethical research standards.

An abstract set of short results from the described (agreed) system is expected to demonstrate very good insights into the face recognition operation. These include tracks in video data from face detection, traits extraction, similarity calculation, and accuracy score. The results obtained constitute a basis, from which we can evaluate the applied methods as shown in figure 5, in terms of their ability to extrapolate faces from video streams. Also, a brief discussion of the consequences of these findings on the research objectives and methodologies, practical application of different audio and video sources as well as data sets is considered, allowing to anticipate the variations in the obtained results. This in-depth explanation delivers universal knowledge of the approaches, problems, and ethics in face detection from camera recordings. Researchers and the above-mentioned practitioners can apply this type of data in formulating evidence-based strategies as they seek to create more ethically structured and accommodating face recognition techniques.

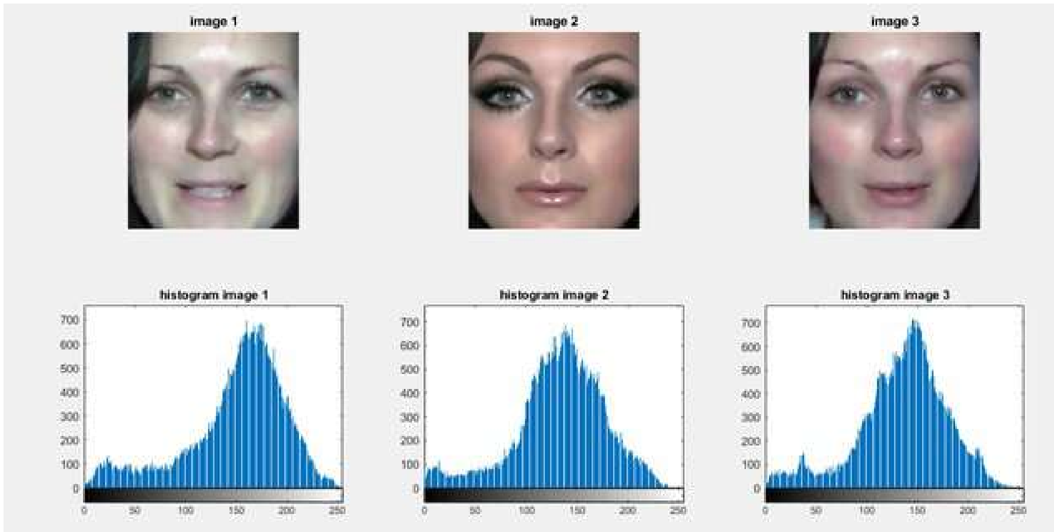

Figure 5. Histogram-based representation of person 1

9 Conclusion

One of the most emerging technologies in Video data analysis is undoubtedly facial recognition. Video Facial recognition technology can be utilized for security surveillance and identity authentication in different applications. Here in this discourse, we have examined the techniques, obstacles, and ethical dynamics playing a crucial role in the unraveling future, an active research area. The method of face recognition employed on video data is a multi-step process that includes face detection, which, in turn, requires methods such as Haar cascades or MTCNN algorithms to detect the faces in video frames. To cover the distinguishable features using deep learning-based models VGG16 and ResNet, feature extraction is done. About the facial feature vectors as input, cosine similarity, cutting-edge metrics like Euclidean distance are used to quantify the feature's likeness.

Integration of Siamese Networks and Triplet Networks which are most accurate for recognition adds to a higher precision level.

In this scope, nevertheless, these sciences have their pain points. Differing with lighting conditions and the geometry of the human body in different poses, temporal consistency, dealing with massive data sets, and real-time processing in videos are the major discriminations that researchers are to tackle. However, since ethics plays a crucial role, ethical considerations cannot be avoided. Data de-identification and request for consent are obligatory if it suits the needs of an individual and the data is identifiable. Security protocols such as data encryption and verification of access identities should be put in place to ensure the safety of data exchanged. Clear disclosure of ethical documents and procedures with data handling also is vital for preserving public trust. The prognosis of real-time face detection using video data has a strong potential to develop in the future as new tools and higher accuracy are introduced. Researchers and practitioners need to choose the right methods and be prepared with valid protocols and regulations for developing ethical and accurate face recognition systems.

With the technological evolution, it becomes paramount to focus on the privacy, transparency, and safety of data that in turn will help unlock the potential of technological innovations to the best of society's interest along with the highest ethical scruples. Implementing this speaks for the necessity of video face recognition as against disclosing personal data and private details.

10 Future work

In the coming time, further research in facial recognition classifiers is likely to orient towards the algorithms so that they can be more accurate and sturdier. We propose methods focusing on upgrading the technique of feature extraction, developing more reliable face detection models, and fine-tuning the distance measure to detect cases of false similarities. Deep learning networks will mainly improve and introduce more advanced face recognition models, while challenging scenarios, such as opacity, ill light, and pose variations are also taken into consideration. Our attention will be focused on building from video streams as close to real-time facial recognition as possible. This will be enabled by optimized algorithms and hardware neural networks' acceleration that will give the desired low-latency recognition needed in applications such as surveillance and access control. Concerns about privacy of the individuals will grow because of the continuous leveraging of facial recognition techniques hence the need to research privacy-preserving facial recognition will be essential. Some of these techniques might include federated learning or secure multi-party computing to be used in SSL to be able to recognize faces while respecting people's right to privacy. Being able to train models on more diverse sets of data with a lot of different demographics and environments will prove to be the key to creating models that do not have any inclusivity or bias issues.

Besides facial recognition, by using other biometric data, like voice recognition or behavioral biometrics, we can build a solid security system and reduce the opportunity to use phishing. Furthermore, high transparency and interpretability of facial recognition systems should be emphasized in areas where the decision-making processes must necessarily be explained. Customer authentication via facial recognition is reality on cutting-edge devices including smartphones and IoT cameras as a growing tendency. Hence, creating algorithms that can manipulate limited resources and resources will be the most vital aspect. As technology advances over time there is going to be increased attention and regulation which could lead to more research works that coincide with the guidelines of what is right and wrong conduct (the ethical guidelines and legal frameworks), so there would be responsible and fair use. Lastly, human-focused applications will prevail as well, for instance, in healthcare for medical cards, in education for students' accounting presence, and commerce for customers' business keeping. Profiling, among others. Inclusivity, accessibility, and resistance to adversarial attacks will continue to be important factors in the continued development of facial recognition technology.

Author Contributions

Muhammad Latif: Conceptualization, Methodology, Software **Mansoor Ebrahim**: Supervision **Abdul Salam Abro**: Visualization, Investigation. **Maaz Ahmed**: Data curation, Writing- Original draft preparation. Software, Validation. **Muhammad Daud Abbasi**: Writing- Reviewing. **Imran Aziz Tunio**: Editing

Compliance with Ethical Standards

It is declared that all authors don't have any conflict of interest. It is also declared that this article does not contain any studies with human participants or animals performed by any of the authors. Furthermore, informed consent was obtained from all individual participants included in the study.

References

[1] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, "Deepface: Closing the gap to human-level performance in face verification," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 1701–1708.

[2] X.-X. Li and R.-H. Liang, “A review for face recognition with occlusion: from subspace regression to deep learning,” *Chinese Journal of Computers*, vol. 41, no. 1, pp. 177–207, 2018.

[3] A. A. Abro, M. S. H. Talpur, and A. K. Jumani, “Natural language processing challenges and issues: A literature review,” *Gazi University Journal of Science*, pp. 1–1, 2023.

[4] A. J. Shepley, “Deep learning for face recognition: a critical analysis,” *arXiv preprint arXiv:1907.12739*, 2019.

[5] S. Kanagamalliga, R. Abishek, B. B. S. Krishna, and P. Vinayagam, “Advancements in real-time face recognition algorithms for enhanced smart video surveillance,” *Procedia Computer Science*, vol. 230, pp. 486–492, 2023.

[6] W. Ali, W. Tian, S. U. Din, D. Iradukunda, and A. A. Khan, “Classical and modern face recognition approaches: a complete review,” *Multimedia tools and applications*, vol. 80, pp. 4825–4880, 2021.

[7] S. Moiz, A. A. Abro, M. Ebrahim, and A. B. Abro, “Unveiling the arsenal of user data protection tools and practices,” in *2024 IEEE 1st Karachi Section Humanitarian Technology Conference (KHI-HTC)*. IEEE, 2024, pp. 1–7.

[8] S. Moiz, A. Majid, A. Basit, M. Ebrahim, A. A. Abro, and M. Naeem, “Security and threat detection through cloud-based wazuh deployment,” in *2024 IEEE 1st Karachi Section Humanitarian Technology Conference (KHI-HTC)*. IEEE, 2024, pp. 1–5.

[9] F. J. M. Shamrat, M. Al Jubair, M. M. Billah, S. Chakraborty, M. Alauddin, and R. Ranjan, “A deep learning approach for face detection using max pooling,” in *2021 5th international conference on trends in electronics and informatics (ICOEI)*. IEEE, 2021, pp. 760–764.

[10] S. Khan, S. Vohra, S. A. Siddique, A. A. Abro, and M. Ebrahim, “A computer vision-based vehicle detection system leveraging deep learning,” in *2024 IEEE 1st Karachi Section Humanitarian Technology Conference (KHI-HTC)*. IEEE, 2024, pp. 1–7.

[11] J. Chai, H. Zeng, A. Li, and E. W. Ngai, “Deep learning in computer vision: A critical review of emerging techniques and application scenarios,” *Machine Learning with Applications*, vol. 6, p. 100134, 2021.

[12] L. Tharaka, D. Vidanagama, and W. Wijayarathne, “Voice command and face motion based activated web browser for differently abled people,” 2022.

[13] A. A. Abro, “Vote-based: Ensemble approach,” *Sakarya University Journal of Science*, vol. 25, no. 3, pp. 858–866, 2021.

[14] A. A. Abro, E. Taşcı, and A. Ugur, “A stacking-based ensemble learning method for outlier detection,” *Balkan Journal of Electrical and Computer Engineering*, vol. 8, no. 2, pp. 181–185, 2020.

[15] S. M. Daniyal, A. Masood, M. Ebrahim, S. H. Adil, and K. Raza, “An improved face recognition method based on convolutional neural network,” *Journal of Independent Studies and Research Computing*, vol. 22, no. 1, pp. 103–110, 2024.

[16] S. M. Daniyal, M. M. Abbasi, D. Hussain, U. Amjad, A. B. Abro, and M. Naeem, “A hybrid approach for simultaneous effective automobile navigation with de and pso,” *VAWKUM Transactions on Computer Sciences*, vol. 12, no. 2, pp. 01–15, 2024.

[17] M. M. Abbasi, S. M. Daniyal, A. A. Abro, D. Hussain, U. Amjad, and N. B. Zahid, “Applying neural networks to predict ventilator demand: A study of pakistan’s healthcare sector,” *VFAST Transactions on Software Engineering*, vol. 12, no. 3, pp. 217–229, 2024.

[18] M. H. Irshad, M. Ebrahim, A. A. Abro, K. Raza, and S. H. Adil, “Detecting shadows in computer vision: A matlab-based approach,” *The Asian Bulletin of Big Data Management*, vol. 4, no. 1, pp. Science–4, 2024.

[19] A. A. Abro, W. A. Sıddıque, M. S. H. Talpur, A. K. Jumani, and E. Yaşar, “A combined approach of base and meta learners for hybrid system,” *Turkish Journal of Engineering*, vol. 7, no. 1, pp. 25–32, 2023.

[20] J.-N. Zaech, “Vision for autonomous systems: From tracking and prediction to quantum computing,” Ph.D. dissertation, ETH Zurich, 2024.

[21] S. Mhammedi, N. Gherabi, H. El Massari, Z. Sabouri, and M. Amnai, "A highly scalable cf recommendation system using ontology and svd-based incremental approach," *Bulletin of Electrical Engineering and Informatics*, vol. 12, no. 6, pp. 3768–3779, 2023.

[22] T. T. Nguyen, M. T. Dang, A. V. Luong, A. W.-C. Liew, T. Liang, and J. McCall, "Multi-label classification via incremental clustering on an evolving data stream," *Pattern Recognition*, vol. 95, pp. 96–113, 2019.

[23] A. K. Shukla, A. Shukla, and R. Singh, "Automatic attendance system based on cnn–lstm and face recognition," *International Journal of Information Technology*, vol. 16, no. 3, pp. 1293–1301, 2024.

[24] G. Srivastava and S. Bag, "Modern-day marketing concepts based on face recognition and neuro-marketing: a review and future research directions," *Benchmarking: An International Journal*, vol. 31, no. 2, pp. 410–438, 2024.

[25] ——, "Modern-day marketing concepts based on face recognition and neuro-marketing: a review and future research directions," *Benchmarking: An International Journal*, vol. 31, no. 2, pp. 410–438, 2024.

[26] X. Zhou and T.-c. Zhu, "Survey of research on face recognition methods based on depth learning," in *Journal of Physics: Conference Series*, vol. 2717, no. 1. IOP Publishing, 2024, p. 012027.

[27] S. S. Khan, D. Sengupta, A. Ghosh, and A. Chaudhuri, "Mtcnn++: A cnn-based face detection algorithm inspired by mtcnn," *The Visual Computer*, vol. 40, no. 2, pp. 899–917, 2024.

[28] D. Tribuana, A. L. Arda *et al.*, "Image preprocessing approaches toward better learning performance with cnn," *Jurnal RESTI (Rekayasa Sistem dan Teknologi Informasi)*, vol. 8, no. 1, pp. 1–9, 2024.

[29] J. Anil, L. P. Suresh, P. Muthukumar, S. K. Veni, P. Asha, and R. Prasad, "Literature survey on face recognition of occluded faces," in *2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT)*, vol. 1. IEEE, 2024, pp. 1930–1937.

[30] A. Khalifa, A. A. Abdelrahman, T. Hempel, and A. Al-Hamadi, "Towards efficient and robust face recognition through attention-integrated multi-level cnn," *Multimedia Tools and Applications*, pp. 1–23, 2024.